



Shahid Singh

Systems Developer

Automation engineer, full-stack & embedded developer. I build systems that eliminate busywork — ticket workflows, browser tools, hardware controllers, and cross-platform apps — all from Newcastle, South Africa.

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Newcastle, South Africa

Experience

Systems Developer — Jenny Internet

Jun 2025 – Present · 1 yr

Newcastle, South Africa

- Architected the entire DFE ticket workflow system reducing manual ticket handling by ~70% across Support, Operations, and Admin teams.
- Built 20+ production workflow templates including Wireless Support, Customer Callout AI, Equipment Orders, Vehicle Management, Overtime, Expenses, Site Rental, and Admin requests.
- Designed guided troubleshooting flows, Quick Navigate, pre-escalation checklists, customer-in-pain detection, and cross-ticket automation pipelines.
- Integrated Rocket.Chat alerting across all departments for real-time operations visibility.
- Created internal wiki documentation, beginner guides, and wireless troubleshooting playbooks used company-wide.
- Developed Chrome extensions (Jenny Screenshot Assistant, Jenny Vault) for agent productivity with OCR, redaction, annotation, and encrypted template storage.

Senior Support Agent — Jenny Internet

Jun 2023 – May 2025 · 2 yrs

Newcastle, South Africa

- Provided advanced wireless ISP support including no-connection, speed, intermittent, and outage troubleshooting.
- Performed signal analysis, traceroutes, tower/sector checks, LAN sync verification, and queue inspection.
- Mentored junior agents and contributed to internal knowledge base and troubleshooting guides.
- Developed deep understanding of customer pain points that later informed workflow automation design as a developer.

College Lecturer & Server Administrator

2020 – 2023 · 3 yrs

Newcastle, South Africa

- Delivered lectures in IT and computer-related subjects to tertiary students.
- Managed and maintained the college server infrastructure including user accounts, backups, and network stability.
- Developed course materials and practical assessments for systems development modules.

Education

Diploma in Systems Development

Boston City Campus, Newcastle, South Africa · 2025 – Present

- Systems Development (HSYD100-1)
- Information Systems Management (HISM100-1)
- Academic Literacy (HALP100-1)

Skills & Technologies

Workflow & Automation

Heimdall Triggers SendActions DFE Templates Ticket Automation Rocket.Chat API

JSON Workflows

Frontend & Browser Extensions

React TypeScript Manifest V3 Vite HTML/CSS/JS Tesseract.js OCR

Canvas API

Mobile & Desktop Apps

Flutter / Dart Riverpod Drift / SQLite Supabase GoRouter

Embedded & Hardware

ESP32 / Arduino BLE HID LVGL TFT_eSPI C++ 3D Printing Web Serial

Backend & AI

Python FastAPI Ollama / vLLM ChromaDB RAG Docker PostgreSQL

OS & DevOps

Linux (Arch/Debian) Git systemd GitHub Actions Cross-compilation ARM64

Key Achievements

- ◆ Reduced manual ticket handling by ~70% across ISP support operations through custom workflow automation.
- ◆ Built a production-ready BLE macro keyboard (SudoDeck) competing with the Elgato Stream Deck — sold in South Africa for R1,099 with full open-source firmware and tools.
- ◆ Created a 249MB offline educational suite (NexCampus) with 16 study tools, code lab, OCR, dictionary, and interactive solar system — zero internet required.
- ◆ Developed an open-source business management suite (SpiceDesk) for small food shops with POS, inventory, CRM, and expense tracking — MIT licensed.
- ◆ Published 6+ open-source projects on GitHub including Chrome extensions, Flutter apps, and embedded firmware.
- ◆ Self-hosted AI workspace (Odysseus) supporting local LLMs, RAG, agents, email triage, and deep research.
- ◆ Built a custom Linux kernel (SudoOS) for ARM64 handheld gaming devices.

Work Projects — Jenny Internet

DFE Ticket Workflow Templates

20+ templates for Support, Operations, and Admin.

Wireless Support 02 Workflow

Guided troubleshooting for all support scenarios.

Quick Navigate

Jump-to-section navigation for long support tickets.

Customer Callout AI

Structured callout tickets with auto-received data.

Callout Automation Logic

Payment handling, arrangement, cancellation flows.

Heimdall Scheduled Triggers

Dealer-change triggers, duplicate prevention.

HOD Escalation Workflows

State save, assign, return to original agent.

Manager Call Requests

Assignment, notification, auto-return.

Rocket.Chat Notifications

Real-time alerts across all departments.

Customer-in-Pain Detection

Identifies repeat-issue customers automatically.

Overtime Logging

Technician tracking with Rocket.Chat updates.

Vehicle & Expense Linked Workflows

Bidirectional ticket updates.

Equipment Order UX & Automation

Order items, billing, compact info blocks.

CPE Installation Workflows

MikroTik scripts, manual override logic.

Admin/System Request Workflows

Custom checklists, visible progress.

Refund & Site Rental Workflows

Suspend/restart, structured payment terms.

Internal Wiki & Documentation

Company-wide reference guides.

Form UX Improvements

Required fields, collapsible sections, guided steps.

Ticket-to-Ticket Automation

Cross-workflow data passing.

Chrome Extensions for Agents

OCR + redaction + vault tools.

Personal Projects

SudoDeck

C++ ESP32 BLE Arduino Python LVGL

Bluetooth macro keyboard — wireless Stream Deck competitor. ESP32+ILI9341 touchscreen, BLE HID, multi-page grid, 30 buttons/page, app launcher, live widgets, themes, Web Serial config. Sold in SA for R1,099.

[GitHub](#) [Website](#)

SpiceDesk

Flutter Dart Supabase Riverpod

Open-source business management suite for bakeries, delis, and food shops. POS, inventory, CRM, expenses, quotes, reports, WhatsApp sharing, invoice PDFs. Free forever, MIT licensed.

[GitHub](#)

Odysseus

Python FastAPI ChromaDB Ollama Docker

Self-hosted AI workspace — chat with local LLMs, agent with MCP tools, deep research, document editing, email triage, calendar, notes, tasks, memory, and 2FA. Privacy-first, runs on your own hardware.

[GitHub](#)

NexCampus

[Python](#)[JavaScript](#)[OCR](#)[PyWebView](#)

Offline educational suite — 249MB with 16 study tools, code lab (Python/JS/Bash), text tools, 262k-word dictionary, periodic table, interactive solar system, OCR, 9 themes. Free, no internet required.

[GitHub](#)

Family Tree App

[Flutter](#)[Dart](#)[SQLite](#)[Riverpod](#)

Interactive offline family tree builder. Multiple family books, tree view with pan/zoom, person profiles, photos, relationships, timeline, statistics, dark/light themes. Cross-platform (Windows, Android, Linux).

[GitHub](#)

Jenny Screenshot Assistant

[React](#)[TypeScript](#)[Vite](#)[OCR](#)[Manifest V3](#)

Production Chrome/Edge extension for screenshot capture, local OCR, regex extraction, redaction, annotation, and ticket note generation. Privacy-first — no uploads by default, optional AI diagnosis.

[GitHub](#)

Jenny Vault

[JavaScript](#)[Chrome API](#)[Encryption](#)

Chrome extension for saving template JSON/text with local encryption. Password-protected vault with search, edit, JSON/tree views, export, and bulk management.

[GitHub](#)

CYD Chrome Controller

[C++](#)[ESP32](#)[Arduino](#)[Python](#)[ydotool](#)

Physical touch interface for browser media control. ESP32 + touchscreen with 9 media zones (prev/play/next/vol/mute/back/reload/forward).

[GitHub](#)

Agent Robin 007

[HTML](#)[CSS](#)[JS](#)[Canvas API](#)

James Bond-themed Father's Day interactive experience. Animated mission briefing, 3-round shooting gallery, score tracking, Bond quotes, downloadable commendation card.

[GitHub](#)

For Akiva, With Love

[HTML](#)[CSS](#)[JS](#)[Canvas](#)[Web Audio](#)

Interactive web experience — a garden of 9 years of memories. Plant seeds, watch flowers bloom with love letters and photos. Aurora sky, fireflies, moonlit scene, ambient music.

[GitHub](#)

SudoOS

[Linux Kernel](#)[ARM64](#)[Bash](#)[Cross-compile](#)

Custom Linux kernel build for ARM64 handheld devices (R36S). Automated build script, custom kernel config, initramfs, ulnitrtd creation.

[GitHub](#)

